

Chapter 4

Phi-Regret and Multicalibration

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The black-box reduction from Φ -regret to online learning due to Gordon, Greenwald and Marks [GGM08] centers around the algorithmic primitive of fixed points. This chapter introduces a different route to the same destination that passes through forecasting [FP26].

At a high level, the moral of this section is the following, also depicted in Figure 1.

One can construct a Φ -regret minimizer starting from a powerful enough *multicalibrated forecaster* of the upcoming utility, and best respond to such forecast. In turn, the task of constructing such a forecaster admits a reduction to online learning in the style of Gordon-Greenwald-Marks, with expected variational inequalities replacing expected fixed points as the nonlinear optimization primitive.

As we will show towards the end of the chapter, this forecasting-based reduction comes with algorithmic benefits that forego the need for complicated semiseparation (Ref Chapter 3). We note however that it is not known whether a forecasting-based approach can yield fast *offline* algorithms for Φ -equilibrium computation (*cf.* Ref Chapter 2).

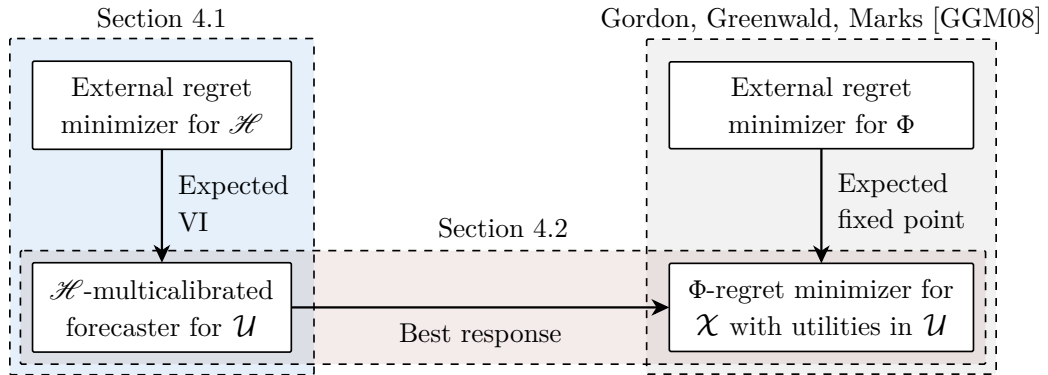


Figure 1: The two forecasting reductions and their relation to the Gordon-Greenwald-Marks framework.

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4.1 A Gordon-Greenwald-Marks result for multicalibration

We start with online multicalibration. In every round t , Nature reveals a context $\mathbf{c}^{(t)} \in \mathcal{C}$. The learner outputs a distribution $D^{(t)}$ over forecasts $\mathbf{p}^{(t)} \in \mathcal{U} \subseteq \mathbb{R}^d$, and Nature then reveals the true utility vector $\mathbf{u}^{(t)} \in \mathcal{U}$. Given test $h : \mathcal{C} \times \mathcal{U} \rightarrow \mathbb{R}^d$, the calibration error against h is

$$\text{MC-Err}^{(T)}(h) := \sum_{t=1}^T \mathbb{E}_{\mathbf{p}^{(t)} \sim D^{(t)}} [\langle h(\mathbf{c}^{(t)}, \mathbf{p}^{(t)}), \mathbf{u}^{(t)} - \mathbf{p}^{(t)} \rangle]. \quad (1)$$

The forecasts are $\mathcal{H}$ -multicalibrated if $\text{MC-Err}^{(T)}(h) = o(T)$ for every $h \in \mathcal{H}$.

As we show in Theorem 4.2, multicalibration admits a black-box reduction to online learning in the style of [Gordon, Greenwald and Marks \[GGM08\]](#). In the case of multicalibration, the nonlinear primitive is *not* fixed points, but rather *expected variational inequalities*, as defined next.¹

Definition 4.1 (Expected variational inequality). Let $S : \mathcal{U} \rightarrow \mathbb{R}^d$ be an operator and let $\varepsilon > 0$. An ε -solution to the expected variational inequality induced by S is a distribution D over $\mathcal{U}$ such that

$$\mathbb{E}_{\mathbf{p} \sim D} [\langle S(\mathbf{p}), \mathbf{u} - \mathbf{p} \rangle] \leq \varepsilon \quad \forall \mathbf{u} \in \mathcal{U}. \quad (2)$$

One way to parse (2) is as a randomized self-consistency condition. The distribution D may place mass on several forecasts; nevertheless, after averaging over that randomness, no possible utility vector $\mathbf{u} \in \mathcal{U}$ has positive correlation with the residual direction $S(\mathbf{p})$ by more than ε . This is exactly what will make the regret analysis telescope. Efficient algorithms for EVIs are known for general compact convex sets under mild oracle access assumptions; for our purposes, the important point is that EVIs are a standalone optimization primitive, just as fixed points were in the Gordon-Greenwald-Marks theorem from Chapter 1.

Now suppose that we have an external-regret minimizer $R_{\mathcal{H}}$ whose decision set is the class of tests $\mathcal{H}$. In each round, it chooses a test $h^{(t)} \in \mathcal{H}$; after the utility vector is revealed, it receives the linear utility

$$g^{(t)}(h) := \mathbb{E}_{\mathbf{p}^{(t)} \sim D^{(t)}} [\langle h(\mathbf{c}^{(t)}, \mathbf{p}^{(t)}), \mathbf{u}^{(t)} - \mathbf{p}^{(t)} \rangle]. \quad (3)$$

The reduction is as follows.

Algorithm: Multicalibration from external regret

Input: An external regret minimizer $R_{\mathcal{H}}$ for the test set $\mathcal{H}$

function NextForecast($\mathbf{c}^{(t)}$):

 Set $h^{(t)} := R_{\mathcal{H}}.\text{NextStrategy}()$

 Let $D^{(t)}$ solve the EVI with $S^{(t)}(\mathbf{p}) := h^{(t)}(\mathbf{c}^{(t)}, \mathbf{p})$:

$$\mathbb{E}_{\mathbf{p}^{(t)} \sim D^{(t)}} [\langle h^{(t)}(\mathbf{c}^{(t)}, \mathbf{p}^{(t)}), \mathbf{u} - \mathbf{p}^{(t)} \rangle] \leq \varepsilon^{(t)} \quad \forall \mathbf{u} \in \mathcal{U}$$

¹Expected variational inequalities have appeared in the literature under different names finish

Algorithm: Multicalibration from external regret

 | **return** $D^{(t)}$

function ObserveUtility($\mathbf{u}^{(t)}$):

 | Feed the linear utility $g^{(t)}$ in (3) to $R_{\mathcal{H}}$

Theorem 4.2 ([FP26]). Let $\text{Reg}_{\mathcal{H}}^{(T)}(h)$ be the external regret of $R_{\mathcal{H}}$ with respect to the sequence of utilities $g^{(1)}, \dots, g^{(T)}$, and let $\text{EVI}^{(T)} := \sum_{t=1}^T \varepsilon^{(t)}$. Then the forecasts produced by the algorithm above satisfy

$$\text{MC-Err}^{(T)}(h) \leq \text{Reg}_{\mathcal{H}}^{(T)}(h) + \text{EVI}^{(T)} \quad \forall h \in \mathcal{H}.$$

Proof. The EVI condition is invoked at the realized utility vector $\mathbf{u}^{(t)}$, so

$$g^{(t)}(h^{(t)}) \leq \varepsilon^{(t)}.$$

Therefore, for any comparator test $h \in \mathcal{H}$,

$$\begin{aligned} \text{MC-Err}^{(T)}(h) &= \sum_{t=1}^T g^{(t)}(h) \\ &\leq \sum_{t=1}^T (g^{(t)}(h) - g^{(t)}(h^{(t)})) + \sum_{t=1}^T \varepsilon^{(t)} \\ &= \text{Reg}_{\mathcal{H}}^{(T)}(h) + \text{EVI}^{(T)}. \end{aligned}$$

Thus, sublinear regret over $\mathcal{H}$, together with $\sum^{(T)} \varepsilon^{(t)} = o(T)$, gives sublinear multicalibration error. $\square$

Necessity of EVIs. In a precise sense, EVIs are also *necessary*. Suppose we already had an efficient $\mathcal{H}$ -multicalibrated forecaster. Fix a context $\mathbf{c} \in \mathcal{C}$ and a test $h \in \mathcal{H}$, and consider the EVI operator $S(\mathbf{p}) = h(\mathbf{c}, \mathbf{p})$. To solve this EVI, run the forecaster for T rounds with the same context $\mathbf{c}^{(t)} = \mathbf{c}$. After it outputs $D^{(t)}$, choose Nature's response as

$$\mathbf{u}^{(t)} \in \arg \max_{\mathbf{u} \in \mathcal{U}} \left\langle \mathbf{u}, \mathbb{E}_{\mathbf{p} \sim D^{(t)}} [S(\mathbf{p})] \right\rangle. \quad (4)$$

For any fixed $\mathbf{u}^* \in \mathcal{U}$, the choice in (4) gives

$$\mathbb{E}_{\mathbf{p} \sim D^{(t)}} [\langle S(\mathbf{p}), \mathbf{u}^{(t)} - \mathbf{p} \rangle] \geq \mathbb{E}_{\mathbf{p} \sim D^{(t)}} [\langle S(\mathbf{p}), \mathbf{u}^* - \mathbf{p} \rangle].$$

Averaging the distributions $D^{(1)}, \dots, D^{(T)}$ uniformly therefore produces a distribution D with

$$\mathbb{E}_{\mathbf{p} \sim D} [\langle S(\mathbf{p}), \mathbf{u}^* - \mathbf{p} \rangle] \leq \frac{\text{MC-Err}^{(T)}(h)}{T} \quad \forall \mathbf{u}^* \in \mathcal{U}.$$

As T grows, multicalibration drives the right-hand side to 0. So an online multicalibrated forecaster gives a black-box EVI solver.

4.2 From Multicalibration to Phi-regret minimization

We now turn the arrow around and use forecasting to make decisions. Let $\mathcal{X} \subseteq \mathbb{R}^d$ be a compact convex action set, and let $\mathcal{U} \subseteq \mathbb{R}^d$ be a compact convex set of possible utility vectors. A deviation class is a family $\Phi \subseteq \{\varphi : \mathcal{X} \rightarrow \mathcal{X}\}$. If the learner outputs distributions $\mu^{(t)}$ over $\mathcal{X}$, its Φ -regret against a deviation $\varphi \in \Phi$ is

$$\Phi\text{Reg}^{(T)}(\varphi) := \sum_{t=1}^T \mathbb{E}_{\mathbf{x}^{(t)} \sim \mu^{(t)}} [\langle \varphi(\mathbf{x}^{(t)}) - \mathbf{x}^{(t)}, \mathbf{u}^{(t)} \rangle]. \quad (5)$$

The reduction asks the forecaster to predict the next utility vector. Given a forecast $\mathbf{p} \in \mathcal{U}$, define the deterministic best response

$$\sigma(\mathbf{p}) \in \arg \max_{\mathbf{x} \in \mathcal{X}} \langle \mathbf{x}, \mathbf{p} \rangle, \quad (6)$$

with ties broken by a fixed rule. If the forecaster outputs a distribution $D^{(t)}$ over forecasts $\mathbf{p}^{(t)}$, the decision maker plays the pushforward distribution $\mu^{(t)}$ induced by $\mathbf{x}^{(t)} = \sigma(\mathbf{p}^{(t)})$.

What tests should the forecaster be calibrated against? For each deviation $\varphi \in \Phi$, define

$$h_\varphi(\mathbf{p}) := \varphi(\sigma(\mathbf{p})) - \sigma(\mathbf{p}), \quad \mathcal{H}_\Phi := \{h_\varphi : \varphi \in \Phi\}. \quad (7)$$

This definition is the whole reduction. The test h_φ measures the direction in utility space in which the deviation φ would improve over the best response to the forecast.

Algorithm: Φ -regret from multicalibration

Input: A forecaster over $\mathcal{U}$ multicalibrated with respect to $\mathcal{H}_\Phi$

NextStrategy():

Query the forecaster and receive a distribution $D^{(t)}$ over forecasts $\mathbf{p}^{(t)} \in \mathcal{U}$
return the pushforward distribution $\mu^{(t)}$ of $\mathbf{x}^{(t)} = \sigma(\mathbf{p}^{(t)})$

ObserveUtility($\mathbf{u}^{(t)}$):

Feed the realized utility vector $\mathbf{u}^{(t)}$ to the forecaster

Theorem 4.3 ([FP26]). If the forecaster in the algorithm above has multicalibration error $\text{MC-Err}^{(T)}$ with respect to $\mathcal{H}_\Phi$, then the decision maker has

$$\Phi\text{Reg}^{(T)}(\varphi) \leq \text{MC-Err}^{(T)}(h_\varphi) \quad \forall \varphi \in \Phi.$$

In particular, sublinear $\mathcal{H}_\Phi$ -multicalibration implies sublinear Φ -regret.

Proof. Fix any $\varphi \in \Phi$. Since $\mu^{(t)}$ is the pushforward of $D^{(t)}$ under $\mathbf{p} \mapsto \sigma(\mathbf{p})$,

$$\begin{aligned} \Phi\text{Reg}^{(T)}(\varphi) &= \sum_{t=1}^T \mathbb{E}_{\mathbf{p}^{(t)} \sim D^{(t)}} [\langle h_\varphi(\mathbf{p}^{(t)}), \mathbf{u}^{(t)} \rangle] \\ &= \sum_{t=1}^T \mathbb{E}_{\mathbf{p}^{(t)} \sim D^{(t)}} [\langle h_\varphi(\mathbf{p}^{(t)}), \mathbf{u}^{(t)} - \mathbf{p}^{(t)} \rangle] + \sum_{t=1}^T \mathbb{E}_{\mathbf{p}^{(t)} \sim D^{(t)}} [\langle h_\varphi(\mathbf{p}^{(t)}), \mathbf{p}^{(t)} \rangle]. \end{aligned}$$

The first term is exactly $\text{MC-Err}^{(T)}(h_\varphi)$. The second term is nonpositive, because

$$\langle h_\varphi(\mathbf{p}), \mathbf{p} \rangle = \langle \varphi(\sigma(\mathbf{p})), \mathbf{p} \rangle - \langle \sigma(\mathbf{p}), \mathbf{p} \rangle \leq 0$$

by the definition of $\sigma(\mathbf{p})$ as a maximizer over $\mathcal{X}$. $\square$

The theorem is useful because the complexity of the required calibration class scales with the complexity of the deviation class. If Φ contains only constant deviations, then $\mathcal{H}_\Phi$ is a class of constant best-response gaps and we recover external regret. If Φ grows toward all maps $\mathcal{X} \rightarrow \mathcal{X}$, then $\mathcal{H}_\Phi$ becomes a strong calibration class and the result approaches the classical connection between calibration and swap regret.

The contextual version is identical. If contexts $\mathbf{c}^{(t)}$ are observed before play and deviations have the form $\varphi : \mathcal{C} \times \mathcal{X} \rightarrow \mathcal{X}$, then we define

$$h_\varphi(\mathbf{c}, \mathbf{p}) := \varphi(\mathbf{c}, \sigma(\mathbf{p})) - \sigma(\mathbf{p}).$$

Multicalibration with respect to these tests gives contextual Φ -regret.

4.2.1 Putting the two reductions together

Combining Theorem 4.2 and Theorem 4.3 gives the promised route from external regret to Φ -regret. Set $\mathcal{H} = \mathcal{H}_\Phi$. Run the multicalibration algorithm as the forecaster inside the best-response algorithm. Then, for every $\varphi \in \Phi$,

$$\Phi\text{Reg}^{(T)}(\varphi) \leq \text{MC-Err}^{(T)}(h_\varphi) \leq \text{Reg}_{\mathcal{H}}^{(T)}(h_\varphi) + \text{EVI}^{(T)}. \quad (8)$$

Thus the burden of Φ -regret minimization shifts to two primitives:

1. no-regret learning over the induced test class $\mathcal{H}_\Phi$; and
2. solving EVIs over the forecast domain $\mathcal{U}$.

The key distinction from the Gordon-Greenwald-Marks path is that we do *not* need to optimize over valid deviations directly. We only need a regret minimizer over tests that contain the maps h_φ . This extra flexibility is what makes the approach especially clean for large structured deviation classes.

4.2.2 Linear deviations

As an illustration, consider linear deviations $\varphi(\mathbf{x}) = \mathbf{M}\mathbf{x}$ over a convex compact set $\mathcal{X} \subseteq \mathbb{R}^d$. The classical GGM approach asks us to understand the geometry of all matrices $\mathbf{M}$ satisfying $\mathbf{M}\mathcal{X} \subseteq \mathcal{X}$, and then compute fixed points of the matrices selected by the regret minimizer. That geometry can be difficult.

The multicalibration route permits a relaxation. From (7),

$$h_{\varphi(\mathbf{p})} = (\mathbf{M} - \mathbf{I})\sigma(\mathbf{p}).$$

If every valid linear endomorphism has spectral norm at most S , then each test above lies in a Frobenius ball of radius on the order of $\sqrt{d}(S + 1)$. Instead of learning over valid endomorphisms, we can learn over the larger class

$$\mathcal{H} := \{\mathbf{p} \mapsto \mathbf{A}\sigma(\mathbf{p}) : \|\mathbf{A}\|_F \leq \rho\},$$

where ρ is chosen large enough to contain all tests h_φ . This larger class is easy: external regret over a Euclidean ball is handled by projected gradient descent. Indeed, the utility sent to the matrix learner at time t is linear:

$$g^{(t)}(\mathbf{A}) = \mathbb{E}_{\mathbf{p}^{(t)} \sim D^{(t)}} [\langle \mathbf{A}\sigma(\mathbf{p}^{(t)}), \mathbf{u}^{(t)} - \mathbf{p}^{(t)} \rangle] = \left\langle \mathbf{A}, \mathbb{E}_{\mathbf{p}^{(t)} \sim D^{(t)}} \left[(\mathbf{u}^{(t)} - \mathbf{p}^{(t)})\sigma(\mathbf{p}^{(t)})^\top \right] \right\rangle_F.$$

Consequently, if $\|\mathbf{x}\|_2 \leq B$ for $\mathbf{x} \in \mathcal{X}$ and $\|\mathbf{u}\|_2 \leq L$ for $\mathbf{u} \in \mathcal{U}$, the standard projected-gradient bound gives $O(BLS\sqrt{dT})$ linear-swap regret, up to constants. This improves the dimension dependence of the recent semi-separation-based route of [Daskalakis, Farina, Fishelson, Pipis and Schneider \[Das+25\]](#), while avoiding semi-separation altogether.

4.2.3 RKHS deviations

The same idea extends beyond finite-dimensional linear classes. Suppose that the deviation functions $\varphi : \mathcal{C} \times \mathcal{X} \rightarrow \mathcal{X}$ lie in a vector-valued reproducing kernel Hilbert space with matrix-valued kernel

$$\Gamma((\mathbf{c}, \mathbf{x}), (\mathbf{c}', \mathbf{x}')) \in \mathbb{R}^{d \times d}.$$

For the contextual reduction, the relevant tests are

$$h_\varphi(\mathbf{c}, \mathbf{p}) = \varphi(\mathbf{c}, \sigma(\mathbf{p})) - \sigma(\mathbf{p}).$$

These tests also lie in an RKHS. Indeed, the first term is represented by composing the deviation kernel with the best-response map, while the second term is represented by the linear kernel on $\sigma(\mathbf{p})$. The resulting kernel is

$$\Gamma'((\mathbf{c}, \mathbf{p}), (\mathbf{c}', \mathbf{p}')) := \Gamma((\mathbf{c}, \sigma(\mathbf{p})), (\mathbf{c}', \sigma(\mathbf{p}'))) + \sigma(\mathbf{p})\sigma(\mathbf{p}')^\top. \quad (9)$$

So, if we have an online learner over the RKHS ball associated with Γ' , Theorem 4.2 supplies a multicalibrated forecaster, and Theorem 4.3 turns it into a no- Φ -regret algorithm. This recovers low-degree polynomial deviations as a special case and also covers infinite-dimensional kernels, such as Gaussian kernels, where the fixed-point route does not have an obvious finite-dimensional endomorphism geometry to exploit.

The moral is that forecasting separates the two hard-looking parts of Φ -regret. Best response converts forecasts to actions; multicalibration certifies that no deviation can systematically exploit the forecast errors; and EVIs provide the self-consistency condition that makes the forecaster black-box reducible to ordinary external regret.

Bibliography for this chapter

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